Raymon Sadhra

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Education

University of California, San Diego — San Diego, CA
M.S. in Computer Science — *Expected Jun 2028*

University of California, Santa Cruz — Santa Cruz, CA
B.S. in Computer Science — *Jun 2026*

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Compiler Construction, Database Systems, Computer Networks, Distributed Computing & Systems, Deep Learning, Machine Learning, Software Engineering

Experience

Capital One — Jun 2026 – Aug 2026
Software Engineer Intern — *Richmond, VA*

- Engineered an **SLA monitoring system** in **Python** and **Databricks**, processing **60K+ records per run**
- Built a **config-driven framework** to generate tenant-specific **SQL** and logic, **cutting setup time by ∼80%**
- Designed a **Delta lookup cache** with parallel writes to avoid historical lookups, **doubling throughput**

Concentrical — Jun 2025 – Aug 2025
Software Engineer Intern — *Camas, WA*

- Engineered a **multi-threaded Node.js repository analysis engine** in **TypeScript**, processing **200K+ LOC**
- Developed **Tree-sitter-based static analyzers** to detect dependencies and cycles across **7,000+ files**

BioMed AI Lab — Mar 2025 – Sep 2025
Machine Learning Engineer / Research Assistant for Dr. Razvan Marinescu — *Santa Cruz, CA*

- Developed a **CGSchNet GNN** to model protein–ligand interactions in coarse-grained molecular dynamics
- Implemented **ECFP and PCA-based spatial embeddings** to encode protein–ligand structure and geometry
- Designed custom loss functions for force and orientation prediction, **improving simulation stability by ∼70%**

Procedural Content Generation Lab — Mar 2025 – Jul 2026
Software Engineer / Project Lead for Dr. Markus Eger — *Santa Cruz, CA*

- Developed a **terrain-adaptive world-building generator** using **Python, NumPy, and Amulet API**
- Designed **terrain-aware A* algorithms** for natural road generation, **retaining ∼96.2%** of the original terrain
- Optimized terrain integration with caching and heightmap preprocessing, **cutting generation time by ∼43%**

Tech4Good Lab — Jan 2025 – Apr 2026
Full-Stack Developer — *Santa Cruz, CA*

- Built a production **full-stack learning platform** using **TypeScript** and **Firebase**, supporting **550+ users**
- Engineered **reactive Firestore data pipelines** with **Cloud Functions** for real-time data synchronization

Projects

RAG Motion | *Python, FastAPI, OpenAI, Chroma, Docker, FFmpeg* — Aug 2026

- Engineered an **end-to-end multimodal video RAG system** for semantic search and grounded Q&A
- Built an **indexing pipeline** to embed transcripts, OCR, captions, and frames for vector retrieval
- Designed an **artifact-driven architecture** with persistent checkpoints, enabling resumable processing

Ghost Runner (CruzHacks 2026 Winner) | *JavaScript, React Native, Expo, Firebase* — Jan 2026

- Engineered a real-time **GPS tracking platform** for live route visualization and activity persistence
- Designed a **replay system** by interpolating timestamped GPS trajectories for sub-second race synchronization

Technical Skills

Languages: Python, C++, C, Java, TypeScript, SQL, JavaScript, HTML, CSS, Assembly
Technologies: AWS, Docker, FastAPI, Node.js, React, Databricks, PySpark, Firebase, MySQL, Git, Unix/Linux
Concepts: Backend Systems, Distributed Systems, System Design, REST APIs, Data Engineering, Concurrent Programming, Performance Optimization, CI/CD, Developer Tooling, Generative AI, AI-Assisted Development